

Question

X is a 1:1 type salt (quaternary compound) with a carbon mass fraction of 4.35% and a boron mass fraction of 43.09%. Treatment of **X** with excess ICl in glyme yields the anion **B**[−] in good yield, which has a boron mass fraction of 13.24%. The reaction of CPh_3^+B^- with $^i\text{Pr}_3\text{SiH}$ in toluene gives pale yellow microcrystals of **C**, and the reaction of **C** with anhydrous liquid hydrogen chloride gives **A**.

Which of the following statements are correct:

1. The quotient of the number of possible structures with the highest symmetry of **B**[−] and its point group order is 0.4.
2. The mass fraction of boron in **A** is approximately 13.22%.
3. Under different reaction conditions, **C**_{Cl}, **C**_{Br}, and **C**_I can be obtained respectively, then the order of the magnitude of the cation-anion interaction in **C** is **C**_{Cl} > **C**_{Br} > **C**_I.
4. Similarly, **A**_{Cl}, **A**_{Br}, and **A**_I exist, and the order of their acidity is **A**_{Cl} > **A**_{Br} > **A**_I.

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **M**

Detailed Explanation

From the mass fraction, it can be known that in **X**, $B : C = 11 : 1$. At this time, one can think of carborane. Since the following is a halogenation process, it can be assumed that the anion is $CB_{11}H_{12}^-$ first. Substituting into the mass fraction calculation, it can be known that the remainder is about 133, i.e., Cs, which is consistent with the conditions of the problem. Therefore, **X** is $Cs[CB_{11}H_{12}]$.

CHECKPOINT

1 PTS

X is $Cs[CB_{11}H_{12}]$

According to the question, **B**[−] must contain C, B, and I elements, and may also contain H element.

Consider from the simplest case, considering that the carborane skeleton remains unchanged, that is, **B**[−] still contains 11 Bs. Since the mass fraction of B is 13.24%, the molar mass of **B**[−] is $898.1 \text{ g} \cdot \text{mol}^{-1}$.

Removing 1 C and 11 Bs, the remainder is $767.2 \text{ g} \cdot \text{mol}^{-1}$, corresponding to 6 Is and 6 Hs. Therefore, the chemical formula of **B**[−] is $CB_{11}H_6I_6^-$.

CHECKPOINT

2 PTS

B[−] is $CB_{11}H_6I_6^-$

When $\text{CPh}_3^+ \text{B}^-$ reacts with $^i\text{Pr}_3\text{SiH}$ in toluene, the negatively charged hydrogen combines with CPh_3^+ , and **C**: $^i\text{Pr}_3\text{Si}[\text{CB}_{11}\text{H}_6\text{I}_6]$ is obtained at the same time. **C** reacts with anhydrous liquid hydrogen chloride to undergo protonation to obtain **A**: $\text{H}[\text{CB}_{11}\text{H}_6\text{I}_6]$.

CHECKPOINT

1 PTS

A: $\text{H}[\text{CB}_{11}\text{H}_6\text{I}_6]$

CHECKPOINT

1 PTS

C: $^i\text{Pr}_3\text{Si}[\text{CB}_{11}\text{H}_6\text{I}_6]$

Statement 1: The highest symmetry requires placing 1 H and 1 I on the C or B that are farthest apart, with the 5 Bs close to H connected to I, and the 5 Bs close to I connected to H; or the 5 Bs close to H connected to H, and the 5 Bs close to I connected to I. This is the C_{5v} point group, a total of $2 \times 2 = 4$ types, and the quotient is $4/10 = 0.4$. Statement 1 is correct.

CHECKPOINT

2 PTS

The quotient of the number of possible structures with the highest symmetry of B^- and the order of its point group is 0.4

Statement 2: $(11 \times 10.81)/898.1 \approx 13.22$, statement 2 is correct.

Statement 3: The order of polarizability is $\text{I} > \text{Br} > \text{Cl}$, so the electron donating ability is $\text{I} > \text{Br} > \text{Cl}$. The interaction between anions and cations is $\text{Si} - \text{X}$ coordination, so $\text{C}_{\text{Cl}} < \text{C}_{\text{Br}} < \text{C}_{\text{I}}$, statement 3 is incorrect.

CHECKPOINT

1 PTS

The order of the magnitude of the interaction between anions and cations is $C_{Cl} > C_{Br} > C_I$

Statement 4: The electron withdrawing ability is $A_{Cl} > A_{Br} > A_I$, so the order of acidity is $A_{Cl} > A_{Br} > A_I$, statement 4 is correct.

CHECKPOINT

1 PTS

The order of acidity is $A_{Cl} > A_{Br} > A_I$

Therefore, choose M